# ANALYSIS DOCUMENTATION

## 1. UNDERSTANDING THE BUSINESS PROBLEM

### Business Context

GloPlus is a global supply chain and logistics services company that provides inventory management, warehousing, transportation and distribution services to businesses across multiple countries, including the United States. The company helps clients move products efficiently from suppliers to customers while maintaining product availability and operational efficiency.

### Business Problem

GloPlus's U.S. operations are facing on-time delivery challenges that may impact customer satisfaction, operational efficiency and profitability. The business needs to identify where delays occur, which customers and operations are most affected, the factors associated with late deliveries and where improvement efforts should be prioritized**.**

### Primary Stakeholder (Supply Chain Operations Manager)

Supply Chain Operations Manager monitors customer fulfillment and delivery performance, identifies operational inefficiencies and improves end-to-end supply chain efficiency.

### Analysis Objective

This analysis will provide the Operations Manager with data-driven insights into delivery performance across customers, locations, carriers and routes, while examining operational factors associated with late deliveries. The findings will support informed decisions on where to prioritize improvement efforts, which areas require further investigation and which operational factors may need to be addressed to improve on-time delivery performance.

### Business and Analytical Questions

| Business Questions | Analytical Questions |
|---|---|
| Which U.S. locations have the most delivery delays? | 1. Which branches, regions or states have the lowest on-time delivery rate?<br>2. Which delivery routes have the highest average delay days? |
| Which customers are most affected by late deliveries? | 1. Which customers have the highest late-delivery rate? |
| Which carriers contribute most to delivery delays? | 1. Which carriers have the lowest on-time delivery rate? |
| What operational factors are linked to late deliveries? | 1. Is longer warehouse processing time associated with a higher late-delivery rate?<br>2. Do long-distance shipments have a higher late-delivery rate? |

### KPIs

| KPI | Why Using This KPI |
|---|---|
| On-Time Delivery Rate | Measures the percentage of shipments delivered by the expected date |
| Average Delay Days | Measures how many days late shipments are, on average |
| Number of Late Deliveries | Shows the volume of late shipments |

| | |
|---|---|
| Average Delivery Time | Measures the average time from shipment to delivery |

**Data Requirements**

| Data Requirement | Required Field | |
|---|---|---|
| Operational Data | • Warehouse Processing Start Date<br>• Warehouse Processing End Date<br>• Dispatch Delay Reason | • Delivery Delay Reason<br>• Shipment Weight<br>• Number of Packages<br>• Service Level<br>• Weather or Traffic Indicator |
| Shipment Data | • Shipment/Order ID<br>• Order Date<br>• Shipment Date<br>• Expected Delivery Date<br>• Actual Delivery Date<br>• Delivery Status | • Branch/Origin<br>• Destination city/state<br>• Route<br>• Carrier Name<br>• Shipping Method<br>• Distance |
| Customer Data | • Customer ID | • Customer Segment |

## 2. DATA COLLECTION

**Dataset Source:** The dataset is a synthetic dataset generated using artificial intelligence to simulate realistic U.S. supply chain and logistics operations. The dataset was designed for educational and portfolio purposes and does not contain real company or customer data. The dataset span from January 1, 2021 to August 31, 2026.

| Tables | Number of rows | Number of columns |
|---|---|---|
| Raw Shipments | 9,900 | 30 |
| Branch Master | 8 | 5 |
| Carrier Master | 16 | 4 |
| Customer Master | 150 | 3 |

## 3. DATA CLEANING

### A. Initial Data Quality Assessment

| Data Quality Issue | Status |
|---|---|
| Missing and blank values | Found |
| Duplicate records | Found |
| Inconsistent text formatting | Found |
| Incorrect data types | Found |
| Mixed date format | Found |
| Extra spaces | Found |
| Invalid and outlier values | Found |

### B. Data Cleaning Steps

| Problem | Action Taken |
|---|---|
| Exact duplicate records | Removed duplicate records, there were 83 records using Excel |
| | **Tool used is Power Query** |
| Incorrect data types | Converted each column to the correct data types |
| Inconsistent Service Level label | Standardized Service Level label by capitalizing each word |
| Inconsistent Branch/Origin names | Standardized Branch/Origin name by capitalizing each word |
| Inconsistent carrier names | Standardized carrier names |
| Inconsistent Delivery Status Label | Standardized Delivery Status label by capitalizing each word |
| Duplicate records after standardization | 9 extra duplicates were removed after standardizing inconsistent labels |
| Null Customer Segment | Filled null customer segment |
| Null Carrier Name | Filled null carrier name |
| Negative Distance, Shipment weight and number of packages | Reviewed negative distance, shipment weight and package values, identified them as invalid and converted them to null |
| Zero Shipment weight and number of packages | Reviewed zero shipment weight and package counts, identified 3 and 4 invalid values respectively and converted them to null |
| Extremely high shipment and number of package values | Reviewed extreme shipment weight and package counts, identified 3 and 2 invalid values respectively and converted them to null |
| Blank values for Weather /Traffic Indicator and Dispatch Delay Reason | Reviewed blank weather/traffic indicators and dispatch delay reasons, identifying 20 and 35 missing values respectively and labelled them as "Unknown." |

### C. Data Quality Decisions

| Issue | Decision |
|---|---|
| Null Customer Segment | Used customer information from the Customer Master table to help fill all the null values using customer ID |
| Null Carrier Name | Used carrier information from the Carrier Master table to help fill all the null values using carrier ID |
| Negative Distance, Shipment weight and number of packages | The negative values were treated as missing values and converted to null |
| Zero Shipment weight and number of packages | The zero values were treated as missing values and converted to null |
| Extremely high shipment and number of package values | The extremely high values were treated as invalid and converted to null |
| Blank Weather/Traffic Indicator and Dispatch Delay Reason | Blank values were categorized as 'Missing' |
| Blank Expected Delivery Date | Maintained as null |
| Blank Actual Delivery Date | Maintained as null |

| | |
|---|---|
| Delivery Status inconsistent with delivery dates | Compared Delivery Status against Expected and Actual Delivery Dates. Recalculated delivery status based on delivery dates |
| Original Delivery Status Quality Flag | Validated the original status against expected and actual delivery dates. Created an Original Status Quality Flag to identify inconsistent and unverifiable records. |
| Delivered Late with missing Actual Date | Identified records where status indicated late delivery but Actual Delivery Date was blank, flagged as missing actual delivery date |
| Returned, Cancelled and In Transit delivery outcomes | Reviewed shipment statuses separately, kept as distinct outcomes |

**D. Created Columns**

| **New Column** | **Purpose** |
|---|---|
| Customer Segment Clean | Fill in all null customer segment making it complete for analysis |
| Carrier Name Clean | Fill in all null carrier names making it complete for analysis |
| Distance Clean | Standardizes distance values making it complete for analysis |
| Shipment Weight Clean | Standardizes shipment weight making it complete for analysis |
| Number of Packages Clean | Standardizes package counts making it complete for analysis |
| Weather/Traffic Indicator Clean | Standardizes missing weather/traffic information by assigning an appropriate missing-value category making it complete for analysis |
| Dispatch Delay Reason Clean | Standardizes missing dispatch-delay information by assigning an appropriate missing-value category making it complete for analysis |
| Delivery Status Clean | To create a consistent delivery outcome based on the delivery dates while preserving Returned, Cancelled and In Transit |
| Original Delivery Status Quality Flag | To help identify the inconsistencies in the original delivery status |

**E. Removed Columns**

| **Removed Column** | **Reason** |
|---|---|
| Customer Segment lookup | Temporary created for null customer segment recovery |
| Carrier Name lookup | Temporary created for null carrier name recovery |
| Customer Segment | Removed after creating the standardized cleaned version |
| Carrier Name | Removed after creating the standardized cleaned version |
| Distance | Removed after creating the cleaned version |
| Shipment Weight | Removed after creating the cleaned version |
| Number of Packages | Removed after creating the cleaned version |
| Weather or Traffic Indicator | Removed after creating the cleaned version |
| Dispatch Delay Reason | Removed after creating the cleaned version |
| Delivery Status | Removed after creating the cleaned version |

| | |
|---|---|
| Carrier Delay Effect | This field was all null and was not needed for analysis |
| Carrier Transit Effect | This field was all null and was not needed for analysis |

## 4. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the cleaned shipment data and identify patterns in customer delivery performance. The analysis focused on overall delivery performance, geographic differences, operational factors, route performance and delay patterns.

**Overall Delivery Performance:** Completed deliveries consisted of 5,278 On Time and 4,285 Late shipments, resulting in an On-Time Delivery Rate of 55.19%. Average late shipments days was 1.45 while the average delivery time was 4.92 days. The remaining included 165 Returned, 115 Cancelled and 14 In Transit, while some records had missing expected or actual delivery dates.

**Geographic Performance:** The Southeast recorded the highest regional on-time rate (58.76%), while the West had the lowest (52.48%). At the state level, Arizona (58.80%) had the highest rate, while Pennsylvania (51.76%) had the lowest. Among branches, Denver Logistic Hub recorded the highest on-time rate (58.33%), while Phoenix Distribution Hub recorded the lowest (50.13%).

**Operational Performance:** Priority had the highest on-time delivery rate for service-level (55.58%), while Road had the lowest shipping-method rate (52.59%). Coastal Express recorded the highest carrier on-time delivery rate (66.63%), compared with Eagle Logistics at 34.41%.

**Route Performance:** Los Angeles, CA to Boise, ID had the highest shipment volume (86 shipments), while Seattle, WA to Spokane, WA had the lowest (14). Denver, CO to San Diego, CA recorded the highest on-time delivery rate (81.82%), while Phoenix, AZ to Miami, FL recorded the lowest (20.69%). Several routes, including Denver, CO to Los Angeles, CA, Denver, CO to Philadelphia, PA and Seattle, WA to Newark, NJ averaged 2 days of delay.

**Delay Patterns:** Carrier Operational Delay was the most frequently recorded delay reason (1,797 late deliveries), followed by Late Dispatch (655). Most late deliveries were 1 day late (2,824), followed by 2 days late (1,098), while 5- and 6-day delays occurred only once each. This indicated that delays were generally short in duration but occurred frequently.

## 5. KPI ANALYSIS

The selected KPIs were used to establish a baseline for customer delivery performance. The On-Time Delivery Rate was 55.19% with a total of 4,285 deliveries were classified as late. Late deliveries averaged 1.45 days beyond the expected delivery date, while the Average Delivery Time was 4.92 days.

The KPIs were then applied to the business questions to compare delivery performance across different areas of the operation. On-Time Delivery Rate were used to evaluate branches and states, Average Delay Days was used to examine routes and Late Delivery Rate were used to evaluate customers, carriers, warehouse processing time and shipment distance.

## 6. BUSINESS QUESTION ANALYSIS

**Business Question 1**: Which U.S. locations have the most delivery delays?

On-time delivery performance was lowest at the Phoenix Distribution Hub (50.13%) and Pennsylvania (51.76%). Route analysis showed the longest delays on Denver → Los Angeles, Denver → Philadelphia, and Seattle → Newark, each averaging 2 days of delay.

**Business Question 2:** Which customers are most affected by late deliveries?

Several customers had significantly low on-time delivery performance. Lighthouse Foods, Maple Manufacturing and Sunrise Solutions had the lowest rates at 14.29%, followed by Summit Health Supplies (22.22%) and Harbor Commercial (25.00%).

These customers performed well below the overall 55.19% U.S. on-time delivery rate, highlighting them as priority areas for investigation into carriers, routes, delay reasons, warehouse processing and shipment distances.

**Business Question 3:** Which carriers or shipping methods contribute most to delivery delays?

Four carriers recorded Late Delivery Rates above 50%, with Eagle Logistics highest at 65.59%, followed by Northstar Carrier Services (58.85%), Summit Transportation (55.22%) and BlueLine Transport (52.57%). These results suggest that carrier performance is a key area for investigation, particularly Eagle Logistics.

**Business Question 4:** What operational factors are linked to late deliveries?

The analysis found no clear relationship between warehouse processing time, shipment distance and late deliveries. Although shipments processed within 2 days had the highest late-delivery rate at 62.50%, shipments processed in 0 days still had a relatively high rate of 41.78%. Similarly, very long-distance shipments had a 47.65% late-delivery rate, while short-distance shipments were slightly higher at 44.85%.

This suggests that processing time and distance alone do not explain late deliveries. Further investigation should focus on carrier performance, dispatch delays, capacity constraints, routes, and delivery-delay reasons.

## 7. DASHBOARD DEVELOPMENT

The Global Plus Delivery Performance Dashboard was developed to provide the Supply Chain Operations Manager with an interactive view of U.S. customer delivery performance. The dashboard brings together key performance indicators and detailed operational analysis to identify where delivery performance is below expectations, which customers were most affected and which operational factors are associated with late deliveries. The dashboard follows a top-down analytical flow, beginning with overall delivery performance and progressing into geographic, route, customer, carrier and operational-level analysis.

Three interactive slicers; Year, Customer Segment and Delivery Delay Reason were utilized which allow users to filter the dashboard and examine delivery performance under different conditions.

Four KPI cards establish the overall delivery-performance baseline: On-Time Delivery Rate (55.19%), Late Deliveries (4,285), Average Delay Days (1.45), and Average Delivery Time (4.92 days). The charts were categorized as follows;

- **Geographic:** Compared On-Time Delivery Rate by Branch/Origin and Destination State to identify delivery-performance gaps across the U.S. network.

- **Route:** Used Average Delay Days by Route to identify routes experiencing longer delays.
- **Customer:** Compared On-Time Delivery Rate by Customer to identify customers with the lowest delivery performance.
- **Carrier:** Compared Late Delivery Rate by Carrier to identify carriers with higher proportions of late deliveries.
- **Warehouse Processing:** Compared Late Delivery Rate across 0, 1, and 2 days of processing to assess whether processing time was associated with late deliveries.
- **Shipment Distance:** Compared Late Delivery Rate across Short, Medium, Long, and Very Long-distance categories to assess whether distance was associated with late deliveries.

Overall, the dashboard translates the cleaned shipment data and KPI analysis into an interactive decision-support tool for the Supply Chain Operations Manager.

## 8. INSIGHTS, RECOMMENDATIONS AND BUSINESS IMPACT

1. **Insight:** GloPlus has a low overall On-Time Delivery Rate of 55.19%, meaning almost half of customer shipments arrive late. Carrier performance varies significantly, with Coastal Express achieving the highest On-Time Delivery Rate at 66.63%, while Eagle Logistics has the lowest at 34.41%, followed by Northstar Carrier Services and Summit Transportation.

   **Recommendation:** Review carrier performance by origin/branch and reallocate suitable shipments to better-performing carriers on comparable routes, such as Coastal Express and Atlas Freight Lines in Los Angeles and RapidRoute Logistics and Pioneer Freight Solutions in Newark. Set a target for underperforming carriers to increase on-time delivery by 10 percentage points within three months, with monthly performance reviews and further shipment reallocation if targets are not met.

   **Business Impact:** Improving carrier performance and delivery reliability can increase customer satisfaction, reduce the risk of customer loss, and help protect future sales.
2. **Insight:** Late shipments are delayed by an average of 1.45 days. Weather causes the longest delays at 1.85 days, followed by Carrier Capacity Constraints (1.62 days) and Late Origin Dispatch (1.60 days). Traffic is also a common cause of delays.

   **Recommendation:** Notify customers early when weather disruptions are expected. During busy periods, move some shipments away from carriers with capacity problems to carriers with better capacity and performance. Also, address delays at the warehouse dispatch stage.

   **Business Impact:** Fewer delays can improve customer satisfaction, reduce “Where Is My Order?” support requests and lower the costs associated with cancellations, refunds and customer service. Better carrier and warehouse performance can also help improve GloPlus’s overall On-Time Delivery Rate.